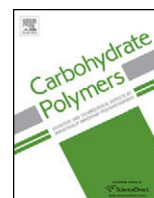




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## CMC stabilized nano silver synthesis, characterization and its antibacterial and synergistic effect with broad spectrum antibiotics

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### ARTICLE INFO

#### Article history:

Received 13 October 2016

Received in revised form

18 November 2016

Accepted 29 November 2016

Available online xxx

#### Keywords:

Silver nanoparticles

CMC

FTIR

XRD

SEM

Antibacterial activity

### ABSTRACT

In the present study silver nanoparticles were synthesized by reduction of AgNO<sub>3</sub> using aqueous CMC solution, which acts as both reducing and capping agent. The formation of AgNO<sub>3</sub> nanoparticles was observed visually by color change and these nanoparticles were characterized through UV–vis spectroscopy, FTIR, XRD, SEM, EDS and AFM. The FTIR peaks observed to be ranging from 3300 to 605 cm<sup>−1</sup>. The AFM image clearly showed the surface morphology of well dispersed nanoparticles. SEM image illustrates the nanoparticles with spherical shape. The crystalline nature of the particles was assured by XRD analysis. The antimicrobial activity of nanoparticles was tested against human bacterial pathogens (*Bacillus cereus*, *Staphylococcus aureus*, *S. epidermidis*, *Escherichia coli*, *Klebsiella pneumoniae*, *Pseudomonas aeruginosa*, *Salmonella typhimurium* & *Vibrio vulnificus*). The bacterial growth was highly inhibited by the nanoparticles. The synergistic effect of nanoparticles in combination with selected broad spectrum antibiotics against the tested bacteria determined strong growth inhibitory activity.

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### 1. Introduction

Nanostructured materials are attracting a great deal of attention due to their potential of achieving specific processes and selectivity, especially in biological and pharmaceutical applications (Li, Li, Wu, Wu, & Li, 2005; Wu et al., 2003). A unique characteristic of synthesized metal nanoparticles is that a change in the absorbance or wavelength gives a measure of the particle size, shape and interparticle properties (Mulvaney, 1996). Among the noble metal nanoparticles, silver nanoparticles (AgNPs) have become the focus of intensive research due to its wide range of applications for many sectors of life and industry. It is due to high surface area and high fraction of surface atoms as compared with bulk silver metal (Kora & Rastogi, 2013). It has many important applications that include spectrally selective coating for solar energy absorption and intercalation material for electrical batteries (Klaus, Joerger, Olsson, & Granqvist, 2001), as optical receptors (Schultz, Smith, Mock, & Schultz, 2000), optoelectronics (Maier et al., 2001), polarizing filters and catalysts in chemical reactions (Kamat, 2002), sensors (Dubas & Pimpan, 2008), biolabelling (Wu et al., 2003),

as antimicrobial agents (Bhainsa & DeSouza, 2006), dental, medical therapeutics and diagnosis (Groneberg, Giersig, Welte, & Pison, 2006), food packaging (Cushen et al., 2012), catheters, textiles etc (Lim, Asharani, & Hande, 2012).

The most common strategy is to protect the nanoparticles with protective agents that can be absorbed on or bind onto the nanoparticles surface, avoiding their agglomeration (Bai et al., 2007). Stabilizers play a crucial role in controlling the size and dispersion stability of particles (Bin Ahmad et al., 2012). Carbohydrates can act not only as stabilizers, but also as a reducing agent, leading to the formation of noble metal nanoparticles. Such conjugated water soluble nanoparticles can have potential antimicrobial applications (Valodkar et al., 2012). Polysaccharides are non-toxic, biocompatible, easily biodegradable, and abundantly present in natural sources. Therefore, many researchers have been used various natural biopolymers for reducing and stabilizing the metal nanoparticles (Kanmani & Lim, 2013). The stabilizing agent used in the present experiment is carboxymethyl cellulose (CMC), which is a well known food grade, non-toxic, biodegradable and biocompatible polysaccharide. CMC is a cellulose derivative with carboxy methyl groups (–CH<sub>2</sub>–COOH) bound to some of the hydroxyl groups of the glucopyranose monomers that make up the cellulose backbone. It is a water soluble carbohydrate polymer, can be produced industrially in a large scale and it has wide range of applications in food and medical related areas. CMC is also used in

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pharmaceuticals as a thickening agent, and in the oil-drilling industry as an ingredient of drilling mud, where it acts as a viscosity modifier and water retention agent.

Human beings are often infected by microorganisms such as bacteria, molds, yeasts and viruses present in their living environment. Because of the emergence and increase in the number of multi drug-resistant microorganisms and also the continuing emphasis on health-care, scientists are trying to develop new effective antimicrobial agents that overcome the resistance of those microorganisms. The emergence of nanoscience and nanotechnology in the last decade represents opportunities for exploring the bactericidal effect of metal nanoparticles. The bactericidal effect of metal particles has been attributed due to their small size and high surface to volume ratio (Morones et al., 2005). Silver nanoparticles (AgNPs) are attractive, because they are non-toxic to the human body at low concentrations and have broad spectrum antibacterial actions (Baker, Pradhan, Pakstis, Pochan, & Shah, 2005).

The present study describes the mechanism of formation of silver nanoparticles using carboxy methyl cellulose (CMC), which has been widely used as a good stabilizer in the synthesis of metal nanoparticles in an aqueous medium (Liu, He, Durham, Zhao, & Roberts, 2008). The present work was conducted to develop cost effective fabrication of spherical silver nanoparticles at ambient temperature and in short reaction time. Also we have investigated the antimicrobial activity of the synthesized silver nanoparticles against both gram positive and gram negative bacteria. The result signifies that silver nanoparticles fabricated using CMC are suitable for formulation of new types of bactericidal materials.

## 2. Experimental

### 2.1. Materials

Silver nitrate (AgNO<sub>3</sub>), Trisodium citrate (C<sub>6</sub>H<sub>5</sub>O<sub>7</sub>Na<sub>3</sub>), Sodium hydroxide (NaOH), Sodium carboxy methyl cellulose (NaCMC), Nutrient agar, Nutrient broth, Mueller Hinton agar, Agar agar, and standard antibiotic discs were purchased from Himedia Private Limited, Mumbai. Milli-Q water was used for the fabrication of nanoparticles and also for preparation of all other aqueous solutions throughout the experiment.

### 2.2. Synthesis of silver nanoparticles using CMC (Vasileva, Donkova, Karadjova, & Dushkin, 2011)

All glassware's used in the experiment were cleaned well using nitric acid and rinsed with double distilled water, followed by acetone for prior to use. The aqueous dispersion of silver nanoparticles was synthesized by the chemical reduction of AgNO<sub>3</sub> using D-glucose as an environmentally benign reducing agent, CMC as a non-toxic stabilizing agent and NaOH as a base reaction catalyst. Freshly prepared aqueous solution of CMC (0.2% [w/v], 32 ml) was stirred for 15 min. Then, 16 ml of  $1 \times 10^{-3}$  M AgNO<sub>3</sub> aqueous solution was added drop by drop under constant stirring. The mixture was stirred for another 10 min to allow the diffusion of metal ions to the stabilizing agent. After that, D-glucose ( $1 \times 10^{-1}$  M, 480  $\mu$ l) was added drop wise under continuous stirring, followed by the addition of NaOH (2.4 ml,  $1 \times 10^{-1}$  M). The entire process was carried out at a constant temperature of  $30 \pm 1$  °C and the reaction was completed within 60 min after the first color appearance.

### 2.3. Characterization of CMC stabilized AgNPs

The CMC stabilized silver nanoparticles were roughly monitored by visual inspection of the reaction solution by color change (Hebeish, El-Rafie, Abdel-Mohdy, Abdel-Halim, & Emam, 2010). UV–vis spectra of the nanoparticles suspension was recorded on

a Shimadzu dual beam spectrophotometer (UV-1650 PC) operated at a resolution of 1 nm at the wavelengths of 200–800 nm. The FT-IR spectrum of nanoparticles was carried on Shimadzu FT-IR spectrophotometer at the range between 4000–550 cm<sup>-1</sup> (Pandey, Goswami, & Nanda, 2012). For comparison, the FT-IR spectrum of CMC stabilizer alone was used as control. The crystallographic analysis was performed by powder X-ray diffraction with the scanning mode on an X'pert PROPAN analytical instrument operated at 40 kV and a current of 30 mA with Cu  $\alpha$  radiation ( $\lambda = 1.54060$  Å). The diffraction intensities were compared with the standard JCPDS files (Valodkar, Bhadoria, Pohnhekar, Mohan, & Thakore, 2010). Morphology of the synthesized silver nanoparticles was observed with the scanning electron microscope (JSM 35 CF JEOL) operated at a resolution of 60 Å at 15 kv (Hebeish, Hashem, Abd El-Hady, & Sharaf, 2013). The quantitative distribution of polysaccharide stabilized silver nanoparticles was investigated by ED's analysis (JSM 35 CF JEOL) in a resolution of 60 Å with the magnification of 5 k (Hebeish et al., 2013). AFM of CMC stabilized silver nanoparticles was measured and characterized using Atomic Force Microscopy (Nanonics imaging MN1000), which measured the atomic range of particles using tapping mode (Sharma, Yngard, and Lin, 2009).

### 2.4. Antibacterial studies of CMC stabilized silver nanoparticles

#### 2.4.1. Bacterial culture

Both gram positive and gram negative human bacterial pathogens such as *Bacillus cereus* (MTCC 619), *Escherichia coli* (MTCC 4296), *Klebsiella pneumoniae* (MTCC 4030), *Pseudomonas aeruginosa* (MTCC 424), *Staphylococcus aureus* (MTCC 3160), *Staphylococcus epidermidis* (MTCC 435), *Vibrio vulnificus* (MTCC 1146) and *Salmonella typhimurium* (MTCC 3216) were procured from the Microbial Type Culture Collection (MTCC), Chandigarh, India. All the cultures were grown on nutrient agar plates and maintained in nutrient agar slants at 4 °C. Overnight cultures in nutrient broth were used throughout the experiment.

#### 2.4.2. Antibacterial assay

The antibacterial activity of CMC stabilized silver nanoparticles was evaluated using agar-well diffusion method (Kaviya, Santhanalakshmi, & Viswanathan, 2011). In brief, 20 ml of each sterile Mueller Hinton Agar (MHA) medium was poured into the sterile petri plates and kept at room temperature for solidification. Then, each pathogenic bacterium was swabbed uniformly on to the individual MHA plates using sterile cotton swabs. Wells of 6 mm diameter were made on MHA plates using sterile cork borer and in each well 100  $\mu$ l of nanoparticles suspension was poured. The plates were incubated at 37 °C for 24 h and the antibacterial activity of nanoparticles was observed through zone of inhibition (in mm) on the plates.

#### 2.4.3. Determination of minimum inhibitory concentration (MIC) and minimum bactericidal concentration (MBC)

The minimum inhibitory concentration was determined based on broth dilution method (Ruparelia, Chatterjee, Duttagupta, & Mukherji, 2008). 5 ml of each nutrient broth containing various concentrations of CMC stabilized silver nanoparticles (10–100  $\mu$ g/ml) were taken in sterile test tubes, and were sonicated for 10 min after adding the nanoparticles to prevent aggregation. Subsequently, the tubes were inoculated with 0.1 ml of freshly prepared bacterial suspension and then incubated overnight at room temperature in an orbital shaker at 200 rpm. After incubation, the MIC of each test strain was measured at 600 nm using a spectrophotometer (SYSTRONICS, Visible-Spectro 105).

The MBC i.e., the lowest concentration of silver nanoparticles that kills 99.9% of the bacteria was also determined through agar plate method (Wang et al., 2012). To evaluate the MBC, 100  $\mu$ l of

sample from each tube without visible growth was transferred into nutrient agar plate and then incubated overnight at room temperature. The concentration of silver nanoparticles causing bactericidal effect was determined based on the absence of colonies on the agar plates (the population in agar plate less than 10 was regarded as no growth).

### 2.5. Synergistic effect of CMC stabilized silver nanoparticles

The synergistic effect of CMC-AgNps was determined using disc diffusion method (Fayaz et al., 2010). To determine the synergistic effect, five standard antibiotic discs such as, amoxycillin (30 mcg), ampicillin (10 mcg), erythromycin (15 mcg), kanamycin (30 mcg) and tetracycline (30 mcg) were selected and all these discs were impregnated individually with 100  $\mu$ l of freshly prepared CMC stabilized silver nanoparticles and were placed on to MHA plates previously loaded with individual test organisms. The above standard antibiotic discs alone were used as positive controls. The plates were incubated overnight at 37 °C. After incubation, the result was recorded by measuring the inhibitory zone diameter (mm).

### 2.6. Assessment of fold area increase and activity index

The fold area increase (Shahverdi, Fakhimi, Shahverdi, & Minaian, 2007) and activity index (Singariya, Kumar, & Mourya, 2012) were calculated by the mean surface area of the inhibition zone generated by an antibiotic alone and in combination with the CMC stabilized silver nanoparticles using the following equations:

$$\text{Foldareaincrease} = b^2 - a^2 / a^2$$

where,

'a' is the zone of inhibition for the antibiotic and

'b' is the antibiotic with CMC stabilized silver nanoparticles.

Activity index (AI) = Inhibition zone of the sample / Inhibition zone of the standard.

### 2.7. Statistical analysis

The data obtained in the present study are expressed as mean  $\pm$  SD and were analysed using two-way ANOVA at 5% level of significance using the computer software STATISTICA 06 Statsoft, Bedford, UK).

## 3. Results and discussion

Carboxy methyl cellulose has been used as protecting agent for the fabrication of silver nanoparticles, but no reports have been found describing the use of this CMC polymer both as reducing and as stabilizing agents in synthesis of silver nanoparticles in aqueous solution at room temperature. In the present study the reaction solution color was changed from colorless to yellow indicated the formation of colloidal CMC. These color change arises because of the excitation of surface plasmon vibrations in the silver nanoparticles (Mulvaney, 1996). The negatively charged solubilized CMC facilitates the attraction of the positively charged silver cations to the polymeric chains followed by reduction with the existing reducing groups.

### 3.1. Characterization studies of CMC stabilized silver nanoparticles

#### 3.1.1. UV-vis spectroscopy

The CMC stabilized silver nanoparticles gave absorbance in the visible region of the electromagnetic spectrum (400–420 nm) due to the surface plasmon resonance (SPR) transition. The absorbance of the CMC stabilized silver nanoparticles is observed at 419 nm (Fig. 1). The UV-vis absorption spectrum of the same sample was taken after 30 days and it has been found that the absorption maxima was shifted to 413 nm, which is very close to the initial peak position indicating almost no agglomeration of CMC-Ag nanopar-

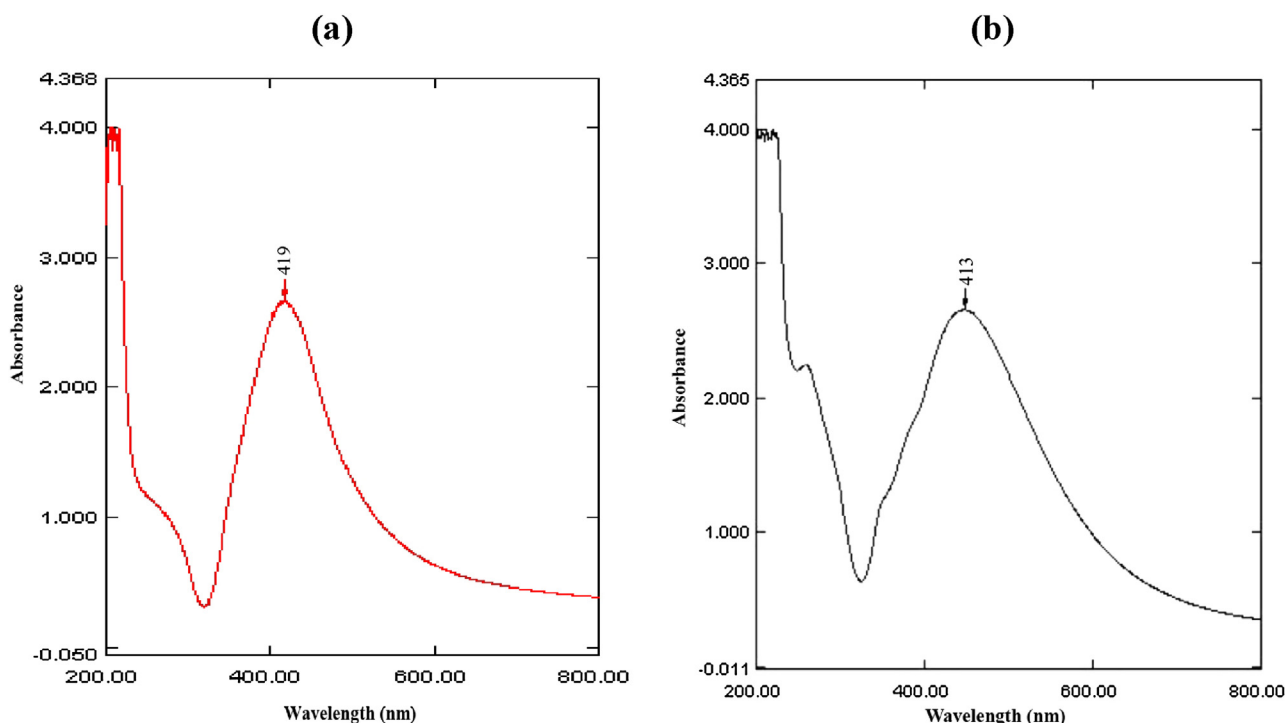


Fig. 1. UV-vis spectroscopy of CMC stabilized silver nanoparticles (a) initial and (b) after 30 days.

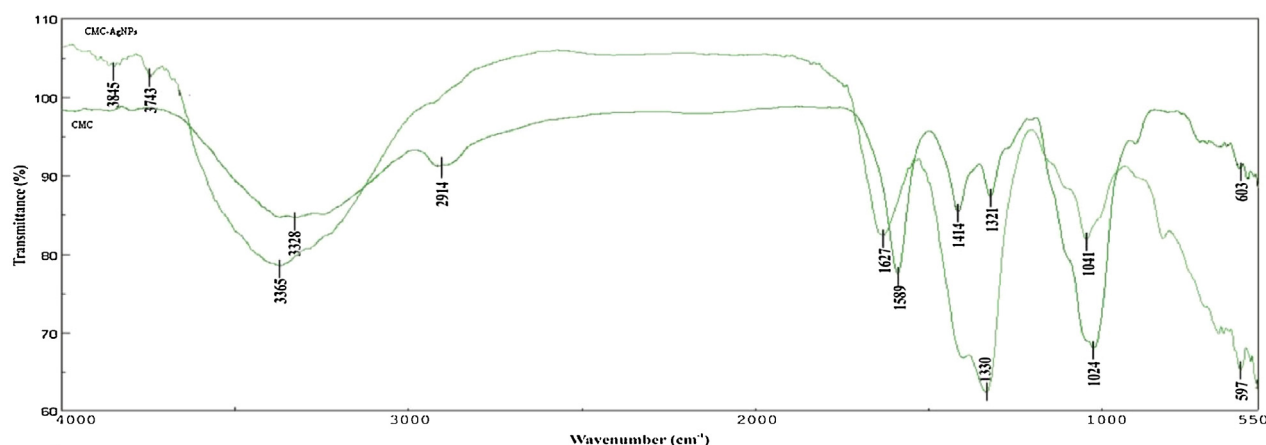


Fig. 2. FT-IR spectrum of CMC and CMC stabilized silver nanoparticles.

ticles. This phenomenon reveals that the silver nanoparticles are well capped with CMC polymer (Hebeish et al., 2010).

### 3.1.2. Fourier transform infra red spectroscopy (FT-IR)

The FT-IR spectrum of CMC-Ag nanoparticles was recorded at 3845, 3743, 3365, 1627, 1414, 1321, 1024 and 603 cm<sup>-1</sup> (Fig. 2). The major peak for CMC was approximately at 3328 cm<sup>-1</sup> due to the stretching frequency of –OH group and a band at 2914 cm<sup>-1</sup> was attributed to C–H stretching vibration. Moreover, the presence of strong absorption band at 1589 cm<sup>-1</sup> corresponds to the coupling of C–O and N–H stretching. The band around 1330 and 1041 cm<sup>-1</sup> represent C–O stretching. The peaks at 1327 and 1262 cm<sup>-1</sup> belong to stretching vibration of C–O and C–C respectively (Hebeish et al., 2010). Whereas, the band appeared at 1064 cm<sup>-1</sup> is characteristic for bending vibration of OH group. Similarly in the present attempt also, the OH group bending vibration was observed at 1024 cm<sup>-1</sup> and C–O stretching at 1321 cm<sup>-1</sup>. Pandey et al. (2012) suggested that the hydroxyl groups in polymer have efficient coordination ability with silver.

### 3.1.3. X-ray diffraction (XRD)

The X-ray diffraction pattern of silver nanoparticles stabilized with CMC showed a number of Bragg's reflections with 2θ values of 38.06°, 44.10°, 64.44°, and 77.20° corresponding to (111), (200), (220), and (311) sets of lattice planes of face centered cubic (fcc) structure of metallic silver ions revealing that the synthesized silver nanoparticles were composed of pure crystalline silver (Fig. 3a). The XRD results obtained in the present experiment which correlate with the earlier report by Das, Das and Das (2013), who examined the XRD peaks at 2θ value of 37.91°, 43.71°, 64.06°, and 76.98°, corresponding to (111), (200), (220), and (311) planes of the face centered cubic of AgNPs respectively. Similarly, four Bragg's diffraction peaks at 2θ value of 38.10°, 44.30°, 64.60°, and 77.50° which could be indexed as (111), (200), (220), and (311) reflections of face centered cubic cc structure of metallic silver (Vasileva et al., 2011).

### 3.1.4. Scanning electron microscopy (SEM)

SEM micrograph of CMC stabilized silver nanoparticles exhibited that the synthesized nanoparticles were spherical in shape (Fig. 3b). Similarly, Valodkar et al. (2010), Chili and Revaprasadu (2008) and Ram Prasad, Elango, Damayanthi and Saranya (2013) in their attempts stated that the CMC stabilized silver nanoparticles were observed to be spherical in shape and also they emphasized the size and shape of these silver nanoparticles depend on many parameters such as choice of reduction technique, concentration of metal precursor, reductant and capping agent used etc.

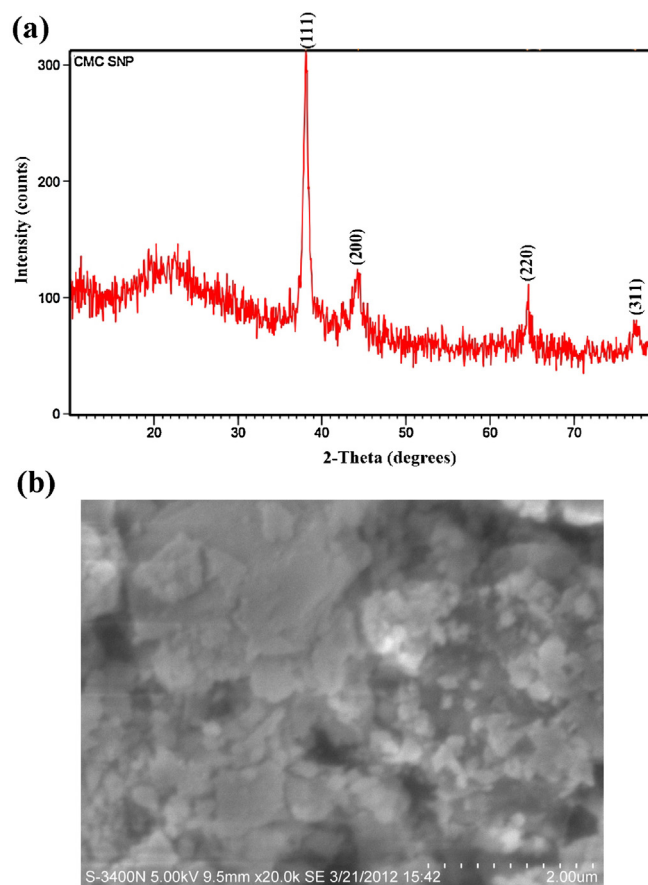


Fig. 3. X-Ray diffraction spectrum (a) and Scanning Electron micrograph (b) of CMC stabilized silver nanoparticles.

### 3.1.5. Energy dispersive spectroscopy (EDS)

The EDS spectrum of CMC-AgNPs showed a strong peak at the energy of 3.0 keV for silver, which confirmed the formation of silver nanoparticles and also some of the weak peaks for Na, C and O were found (Fig. 4a). In accordance with these, Shukla, Singh, Reddy and Jha (2012) observed the same type of elemental Ag signal in the EDX spectrum of silver nanoparticles with the strongest peaks at 3.0 keV, confirming the presence of elemental Ag in the nanoparticles. Furthermore, the EDX pattern of CMC stabilized sil-

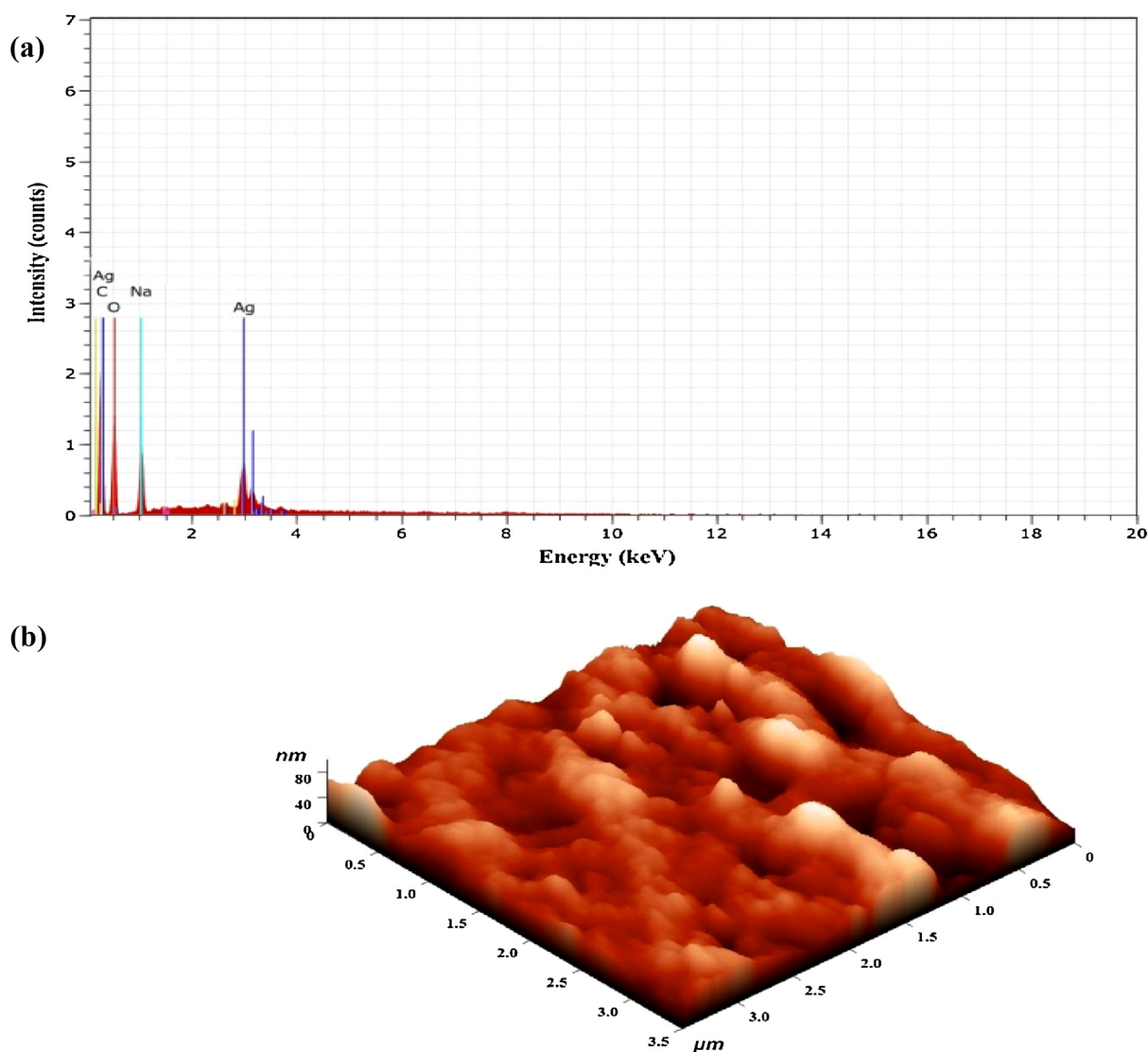


Fig. 4. Energy dispersive spectrum (a) and AFM 3-D image (b) of CMC stabilized silver nanoparticles.

ver nanoparticles showed the presence of strong signal in the silver region at 3.0 keV (Hebeish et al., 2013).

### 3.1.6. Atomic force microscopy (AFM)

The atomic force microscopy result displayed the size, shape, surface morphology and distribution of CMC stabilized silver nanoparticles (Fig. 4b). Surface topology of CMC stabilized silver nanoparticles was spherical in shape and also indicated that the particles were well separated showing no agglomeration in the two dimensional micrograph. The 3D image of the stabilized nanoparticles showed sharp peaks with lower to higher sizes of the nanoparticles. In accordance with these, the bright spots on the surface topology of AFM micrograph indicated the presence of spherical shape silver nanoparticles (Prasad & Swamy, 2013). In the present study the size of CMC stabilized silver nanoparticles was ranged from 5 to 15 nm. The size and shape of the nanoparticles depends on the concentration and type of reducing as well as stabilizing agents used in the synthesis (Sharma et al., 2009).

## 3.2. Antibacterial studies of CMC stabilized silver nanoparticles

### 3.2.1. Antibacterial assay

Table 1 illustrates the bactericidal effect of silver nanoparticles with CMC as stabilizing agent against selected both gram

Table 1

Zone of inhibition (mm) of silver nanoparticles (with and without stabilizers) against selected human bacterial pathogens.

| Bacterial Pathogens   | Zone of Inhibition (mm)     |                            |
|-----------------------|-----------------------------|----------------------------|
|                       | CMC-AgNPs (With stabilizer) | AgNPs (Without stabilizer) |
| Gram-positive         |                             |                            |
| <i>B. cereus</i>      | 20.16 ± 1.12                | 10.60 ± 0.48               |
| <i>S. aureus</i>      | 28.23 ± 1.46                | 11.43 ± 0.68               |
| <i>S. epidermidis</i> | 25.03 ± 1.82                | 10.57 ± 0.37               |
| Gram-negative         |                             |                            |
| <i>E. coli</i>        | 15.30 ± 0.92                | 12.40 ± 0.83               |
| <i>K. pneumoniae</i>  | 30.13 ± 1.48                | 13.30 ± 0.94               |
| <i>P. aeruginosa</i>  | 20.40 ± 1.66                | 14.17 ± 0.98               |
| <i>V. vulnificus</i>  | 16.23 ± 0.88                | 11.00 ± 0.67               |
| <i>S. typhimurium</i> | 18.17 ± 1.13                | 10.03 ± 0.75               |

Each value is the mean ± SD of triplicate analysis.

positive and gram negative human bacterial pathogens. The sensitivity of bacteria towards silver nanoparticles was found to be varied depending on the strains used. Maximum zone of inhibition (30.13 ± 1.48 mm) was observed for CMC-AgNPs against *K. pneumoniae*, however, it exhibited minimum inhibition zone of 15.30 ± 0.92 mm against *E. coli*. There are two possible mecha-

**Table 2**

Determination of MIC and MBC concentration of fabricated silver nanoparticles against selected gram-positive and gram-negative bacteria.

| Bacterial Pathogens   | CMC-AgNPs                |                          |               |
|-----------------------|--------------------------|--------------------------|---------------|
|                       | MIC ( $\mu\text{g/ml}$ ) | MBC ( $\mu\text{g/ml}$ ) | MBC/MIC ratio |
| Gram-positive         |                          |                          |               |
| <i>B. cereus</i>      | 60                       | 60                       | 1.00          |
| <i>S. aureus</i>      | 60                       | 60                       | 1.00          |
| <i>S. epidermidis</i> | 70                       | 70                       | 1.00          |
| Gram-negative         |                          |                          |               |
| <i>E. coli</i>        | 40                       | 50                       | 1.25          |
| <i>K. pneumoniae</i>  | 60                       | 60                       | 1.00          |
| <i>P. aeruginosa</i>  | 50                       | 50                       | 1.00          |
| <i>V. vulnificus</i>  | 60                       | 70                       | 1.17          |
| <i>S. typhimurium</i> | 40                       | 60                       | 1.50          |

nisms for the inhibitory effect of silver nanoparticles on bacterial pathogens (Sondi & Salopek-Sondi, 2004). First, is the electrostatic attraction between the negatively charged cell membrane of the microorganisms and the positively charged silver ions. Second, is the formation of pits on the cell wall of bacteria related to silver ions concentration. The silver nanoparticles with positive charge have greater affinity towards negatively charged bacterium by electrostatic interaction invoking an important determination of the biocidal activity (Stoimenov, Klinger, Marchin, & Klabunde, 2002). Due to higher negative charge on the cell surface, interaction between gram-negative bacteria and positive charge on nanoparticles were definitely stronger than that of gram-positive bacteria (Das et al., 2013). The positive charge containing AgNPs are believed to bind with negative charge containing bacterial cell membranes and disrupt cell walls as well as surface proteins, finally leading to cell death (Sui et al., 2006).

### 3.2.2. Minimum inhibitory concentration (MIC) and minimum bactericidal concentration (MBC) of silver nanoparticles

The experimental result on CMC-AgNPs exhibited that it produced bacteriostatic as well as bactericidal effect against *E. coli* at lower concentrations of 40 and 50  $\mu\text{g/ml}$  (Table 2). Based on the observed data, MIC and MBC values of stabilized silver nanoparticles exerted a bactericidal effect against all the tested bacterial pathogens, because the MBC/MIC ratio was greater than 1 and there was no significant difference between the bactericidal effects of silver nanoparticles on all pathogens. If the MBC/MIC ratio of silver nanoparticles to be  $\leq 4$ , it could be considered as a potent bactericidal agent against the tested organisms (Dugal & Chakraborty, 2013). In accordance with this, in the present study also, the obtained MBC/MIC ratio was lesser than 4, hence, the fabricated silver nanoparticles can also be considered as an effective antibacterial agent towards both gram-positive and gram-negative bacterial strains. The bactericidal effect of AgNPs could be attributed to either interaction with the membrane surface or to their penetration inside the bacteria (Morones et al., 2005). Inside a bacterium, nanoparticles can interact with DNA, thus failing its ability to replicate, which may lead to cell death (Raja, Suriya, Sekar, & Rajasekaran, 2012).

### 3.2.3. Synergistic effect of CMC stabilized silver nanoparticles

The result on synergistic effect of CMC stabilized silver nanoparticles is given in Tables 3 and 4. It revealed that the calculated activity index value of AgNPs was higher than 1 ( $AI > 1$ ), it means that all the bacterial strains used in the present experiment showed very strong inhibitory action than the broad spectrum antibiotics. The maximum activity index value ( $2.80 \pm 0.01$ ) was obtained

against *S. aureus* using CMC-AgNPs with ampicillin as standard antibiotic, followed by Kanamycin ( $2.15 \pm 0.02$ ). While using gram negative organisms, *K. pneumoniae* showed greater activity index ( $3.75 \pm 0.03$ ) by CMC-AgNPs in combination with ampicillin as reference antibiotic than erythromycin. Earlier, Panacek et al. (2006) reported that the interactions of AgNPs with bacteria are dependent on the size and surface area of the nanoparticles. Smaller AgNPs having larger surface area available for interaction would give more bactericidal effect than the larger AgNPs. The antimicrobial activity of AgNPs increases significantly with decreasing size of the nanoparticles, which could be due to the fact that smaller sized particles have larger surface areas for releasing silver ions and also have strong protein binding capability (Kanmani & Lim, 2013). In addition, smaller sized nanoparticles can easily pass through the bacterial membrane pores resulting in strong inhibitory effect. This may due to the size of nanoparticles is lesser than the pore size of bacteria. Thus they have a unique property of crossing the cell membrane without any hindrance (Sunita, Suresh, Madhav, & Anjali, 2011).

Fold area increase of CMC stabilized silver nanoparticles with reference to standard antibiotics against selected human bacterial pathogens as noticed by an increase in diameter of inhibition zones. Among the tested gram positive strains, *S. aureus* exhibited highest increase in fold area ( $3.41 \pm 0.01$ ) with ampicillin as reference antibiotic, followed by *S. epidermidis* ( $1.56 \pm 0.02$ ) using CMC-AgNPs with kanamycin antibiotic. Among the tested gram negative strains, *V. vulnificus* exhibited highest increase in fold area ( $7.27 \pm 0.07$ ) using CMC-AgNPs with ampicillin as reference antibiotic. Earlier Kvitek et al. (2008) reported that *P. aeruginosa* exhibited antibacterial activity in combination with silver nanoparticles, whereas for the antibiotic alone no activity was observed. The reason for the reduced size of inhibition zone for combinations as compared to individual antibiotics may be attributed to the improper dispersion of the sample in the petri dish. It is also observed that the inhibition zone size is almost equal against both bacteria for both the antibiotics suggesting existence of same chelation effect between the silver nanoparticles and antibiotics. The mechanism of the bactericidal effect of Ag NPs can be attributed to the fact that they possibly attach to the surface of the cell membrane disturbing permeability and respiration functions of the cell.

The enhanced surface area to volume ratio of nanoparticles is responsible for the overall mechanism of interference with the cellular machinery, whose exact mechanism remains unknown. It is generally recognized that AgNPs may attach to the microbial cell wall and thus exert a toxic effect by disturbing cell wall permeability. The nanoparticles may also penetrate the cell and affect cellular respiration through inactivating the essential enzymes by forming complexes with the catalytic sulfur of thiol groups in cysteine residues and through the production of toxic radicals such as superoxide, hydrogen peroxide and hydroxyl ions (Feng et al., 2000). In addition to their direct antibacterial activity, nanoparticles are also known to disrupt biofilm formation (Kalishwaralal, Barathmanikant, Pandian, Deepak, & Gurunathan, 2010).

The chemical interactions between the nanoparticles and bacterial cells may also be responsible for the disruption of cell membranes, and it is possible that an induced reductive decomposition of functional groups in the proteins and lipopolysaccharides of the outer membranes (Syed, Manzoor, Shah, & Bukhari, 2010). Earlier, Zhang & Chen (2009) stated that nanomaterials release ions, which react with the thiol group (-SH) of the proteins present on the bacterial surface. Such proteins protrude through the bacterial cell membrane, allowing the transport of nutrients through the cell wall. Nanomaterials inactivate the proteins, decreasing the membrane permeability and eventually causing cellular death.

**Table 3**

Synergistic effect of selected standard broad spectrum antibiotics with and without CMC stabilized silver nanoparticles against selected gram positive human bacterial pathogens.

| Antibiotics           | Gram positive pathogens | Zone of Inhibition (mm) |                              | Zone of difference (mm) (b-a) | Activity Index (b/a) | Fold area increase (b <sup>2</sup> -a <sup>2</sup> /a <sup>2</sup> ) |
|-----------------------|-------------------------|-------------------------|------------------------------|-------------------------------|----------------------|----------------------------------------------------------------------|
|                       |                         | Antibiotic Alone (a)    | CMC –AgNPs + Antibiotics (b) |                               |                      |                                                                      |
| Amoxycillin (30 mcg)  | <i>B. cereus</i>        | 15.67 ± 0.48            | 20.80 ± 1.36                 | 5.13 ± 0.02                   | 1.25 ± 0.01          | 0.72 ± 0.01                                                          |
|                       | <i>S. aureus</i>        | 17.67 ± 1.12            | 25.13 ± 1.82                 | 7.46 ± 0.04                   | 1.56 ± 0.01          | 0.93 ± 0.02                                                          |
|                       | <i>S. epidermidis</i>   | 17.16 ± 1.23            | 24.93 ± 1.64                 | 7.77 ± 0.07                   | 1.47 ± 0.02          | 1.16 ± 0.01                                                          |
| Ampicillin (10 mcg)   | <i>B. cereus</i>        | 10.67 ± 0.94            | 16.10 ± 1.13                 | 5.43 ± 0.05                   | 1.82 ± 0.03          | 1.12 ± 0.02                                                          |
|                       | <i>S. aureus</i>        | 10.00 ± 0.50            | 21.40 ± 1.67                 | 11.48 ± 0.68                  | 2.80 ± 0.01          | 3.41 ± 0.01                                                          |
|                       | <i>S. epidermidis</i>   | 14.90 ± 0.67            | 18.37 ± 1.23                 | 3.47 ± 0.01                   | 1.67 ± 0.02          | 0.44 ± 0.01                                                          |
| Erythromycin (15 mcg) | <i>B. cereus</i>        | 12.00 ± 1.00            | 14.67 ± 0.87                 | 2.68 ± 0.01                   | 1.67 ± 0.02          | 0.56 ± 0.05                                                          |
|                       | <i>S. aureus</i>        | 13.00 ± 1.22            | 18.77 ± 1.24                 | 5.78 ± 0.02                   | 2.15 ± 0.01          | 1.14 ± 0.01                                                          |
|                       | <i>S. epidermidis</i>   | 15.33 ± 1.37            | 20.67 ± 1.48                 | 5.36 ± 0.03                   | 1.67 ± 0.01          | 0.96 ± 0.02                                                          |
| Kanamycin (30 mcg)    | <i>B. cereus</i>        | 16.67 ± 1.48            | 22.20 ± 1.18                 | 5.53 ± 0.05                   | 1.18 ± 0.01          | 0.68 ± 0.05                                                          |
|                       | <i>S. aureus</i>        | 13.00 ± 0.82            | 19.17 ± 1.18                 | 6.18 ± 0.08                   | 2.15 ± 0.02          | 1.14 ± 0.01                                                          |
|                       | <i>S. epidermidis</i>   | 14.87 ± 0.96            | 23.67 ± 1.28                 | 8.68 ± 0.42                   | 1.67 ± 0.01          | 1.56 ± 0.02                                                          |
| Tetracycline (30 mcg) | <i>B. cereus</i>        | 18.66 ± 1.24            | 19.87 ± 1.14                 | 1.22 ± 0.01                   | 1.05 ± 0.01          | 0.11 ± 0.01                                                          |
|                       | <i>S. aureus</i>        | 17.67 ± 1.36            | 24.17 ± 1.82                 | 6.46 ± 0.05                   | 1.55 ± 0.02          | 0.78 ± 0.07                                                          |
|                       | <i>S. epidermidis</i>   | 17.00 ± 1.34            | 25.90 ± 1.67                 | 8.90 ± 0.09                   | 1.47 ± 0.01          | 1.34 ± 0.01                                                          |

Each value is the mean ± SD of five individual replicates.

**Table 4**

Synergistic effect of selected standard broad spectrum antibiotics with and without CMC stabilized silver nanoparticles against selected gram negative human bacterial pathogens.

| Antibiotics           | Gram negative pathogens | Zone of Inhibition (mm) |                             | zone of difference (mm) (b-a) | Activity index (b/a) | Fold area increase (b <sup>2</sup> -a <sup>2</sup> /a <sup>2</sup> ) |
|-----------------------|-------------------------|-------------------------|-----------------------------|-------------------------------|----------------------|----------------------------------------------------------------------|
|                       |                         | Antibiotic Alone (a)    | CMC-AgNPs + Antibiotics (b) |                               |                      |                                                                      |
| Amoxycillin (30 mcg)  | <i>E. coli</i>          | 13.80 ± 0.88            | 20.07 ± 1.69                | 6.27 ± 0.04                   | 1.07 ± 0.01          | 1.04 ± 0.01                                                          |
|                       | <i>K. pneumoniae</i>    | 17.00 ± 1.06            | 23.03 ± 1.85                | 6.03 ± 0.08                   | 1.76 ± 0.01          | 0.83 ± 0.01                                                          |
|                       | <i>P. aeruginosa</i>    | 12.00 ± 0.64            | 18.93 ± 1.35                | 6.93 ± 0.06                   | 1.67 ± 0.02          | 1.52 ± 0.02                                                          |
|                       | <i>S. typhimurium</i>   | 19.77 ± 1.33            | 27.90 ± 1.08                | 8.13 ± 0.09                   | 0.90 ± 0.02          | 0.96 ± 0.01                                                          |
|                       | <i>V. vulnificus</i>    | 08.47 ± 0.46            | 21.93 ± 1.25                | 13.47 ± 0.80                  | 1.98 ± 0.02          | 6.56 ± 0.06                                                          |
| Ampicillin (10 mcg)   | <i>E. coli</i>          | 09.37 ± 0.53            | 18.20 ± 1.28                | 8.84 ± 0.61                   | 1.67 ± 0.01          | 3.00 ± 0.03                                                          |
|                       | <i>K. pneumoniae</i>    | 08.00 ± 0.64            | 17.00 ± 1.07                | 9.02 ± 0.83                   | 3.75 ± 0.03          | 3.52 ± 0.03                                                          |
|                       | <i>P. aeruginosa</i>    | 09.00 ± 0.81            | 15.90 ± 1.08                | 6.87 ± 0.08                   | 2.30 ± 0.02          | 2.16 ± 0.02                                                          |
|                       | <i>S. typhimurium</i>   | 12.00 ± 0.67            | 27.07 ± 1.86                | 15.07 ± 0.96                  | 1.49 ± 0.01          | 4.06 ± 0.04                                                          |
|                       | <i>V. vulnificus</i>    | 08.00 ± 0.43            | 22.96 ± 1.75                | 14.97 ± 0.81                  | 2.30 ± 0.02          | 7.27 ± 0.07                                                          |
| Erythromycin (15 mcg) | <i>E. coli</i>          | 13.00 ± 0.88            | 27.07 ± 1.98                | 14.08 ± 0.87                  | 1.15 ± 0.01          | 3.29 ± 0.03                                                          |
|                       | <i>K. pneumoniae</i>    | 12.33 ± 0.47            | 19.20 ± 1.22                | 6.87 ± 0.31                   | 2.51 ± 0.02          | 1.51 ± 0.01                                                          |
|                       | <i>P. aeruginosa</i>    | 10.00 ± 0.87            | 26.67 ± 1.83                | 16.67 ± 0.81                  | 2.01 ± 0.02          | 6.29 ± 0.06                                                          |
|                       | <i>S. typhimurium</i>   | 15.00 ± 1.04            | 21.53 ± 1.42                | 6.53 ± 0.61                   | 1.21 ± 0.01          | 1.15 ± 0.01                                                          |
|                       | <i>V. vulnificus</i>    | 24.70 ± 1.43            | 27.93 ± 1.69                | 3.23 ± 0.02                   | 1.56 ± 0.01          | 0.25 ± 0.02                                                          |
| Kanamycin (30 mcg)    | <i>E. coli</i>          | 21.00 ± 1.82            | 32.37 ± 2.09                | 11.36 ± 0.81                  | 0.71 ± 0.01          | 1.32 ± 0.01                                                          |
|                       | <i>K. pneumoniae</i>    | 18.70 ± 1.43            | 24.47 ± 1.16                | 5.77 ± 0.31                   | 1.61 ± 0.01          | 0.58 ± 0.05                                                          |
|                       | <i>P. aeruginosa</i>    | 15.50 ± 0.92            | 26.10 ± 1.89                | 10.65 ± 0.63                  | 1.25 ± 0.01          | 1.64 ± 0.02                                                          |
|                       | <i>S. typhimurium</i>   | 16.67 ± 1.27            | 23.13 ± 1.67                | 6.46 ± 0.31                   | 1.06 ± 0.01          | 0.83 ± 0.01                                                          |
|                       | <i>V. vulnificus</i>    | 19.67 ± 1.23            | 21.77 ± 1.20                | 2.15 ± 0.02                   | 0.80 ± 0.01          | 0.21 ± 0.02                                                          |
| Tetracycline (30 mcg) | <i>E. coli</i>          | 14.83 ± 0.84            | 20.97 ± 1.65                | 6.14 ± 0.61                   | 1.10 ± 0.01          | 0.96 ± 0.09                                                          |
|                       | <i>K. pneumoniae</i>    | 17.80 ± 1.28            | 21.80 ± 1.76                | 4.13 ± 0.31                   | 1.67 ± 0.02          | 0.49 ± 0.04                                                          |
|                       | <i>P. aeruginosa</i>    | 18.40 ± 1.56            | 25.23 ± 1.93                | 6.83 ± 0.41                   | 1.11 ± 0.01          | 0.93 ± 0.09                                                          |
|                       | <i>S. typhimurium</i>   | 11.33 ± 0.94            | 17.17 ± 1.17                | 5.84 ± 0.34                   | 1.38 ± 0.01          | 1.38 ± 0.01                                                          |
|                       | <i>V. vulnificus</i>    | 17.90 ± 1.22            | 21.07 ± 1.98                | 3.17 ± 0.02                   | 0.89 ± 0.01          | 0.35 ± 0.03                                                          |

Each value is the mean ± SD of five individual replicates.

#### 4. Conclusion

Nanobiotechnology is an upcoming and developing field with potential application of human welfare owing to its small size and volume ratio to fight against antibiotic resistant pathogens. Silver nanoparticles have received intensive interest because of their various applications in the medical field. The CMC stabilized silver nanoparticles have better bactericidal effect due to an increased surface area. Results from the antibacterial study suggested that silver nanoparticles prepared in a simple and rapid way, may be suitable for the formulation of new types of antibacterial agent and may solve the problems of the emergence and spread

of *in vitro* antibiotic resistance. Hence, the findings in the present study might provide an emerging strategy to combat disease resistant pathogens in future.

#### Acknowledgements

The authors are grateful for the financial support provided by University Grants Commission (Major Project F.No. 39/630/2010 [SR]) and thank V.H.N.S.N. College (Autonomous) Managing Board, Virudhunagar for providing facilities, Alagappa University, CECRI, Karaikudi and Central instrumentation facility Pondicherry University, Pondicherry for technical assistance.

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